

Experiment 4

Aim:

Creation of an account in MetaMask and transfer of cryptocurrency from one account to another using the Sepolia Testnet.

Theory:

1. What is MetaMask?

MetaMask is a non-custodial cryptocurrency wallet available as a browser extension and mobile application. It allows users to create and manage Ethereum-compatible accounts, store cryptocurrency, and interact with blockchain networks and decentralized applications. Since it is non-custodial, the user's private keys and recovery information remain under the user's control. The wallet address can be shared with others for receiving cryptocurrency, while the private key and Secret Recovery Phrase must always be kept confidential.

2. Account Creation in MetaMask

When a new MetaMask account is created, a unique cryptographic account is generated. The account has a public wallet address and a private key. The public address is used to receive funds and identify the account on the blockchain, whereas the private key is used to authorize and sign transactions. MetaMask can also manage multiple accounts from the same wallet interface, allowing users to perform transactions between different accounts.

3. Testnet and SepoliaETH

A **testnet** is a blockchain network used for development, testing, and learning without using real cryptocurrency. In this experiment, the **Sepolia Testnet** is used. SepoliaETH is test cryptocurrency that can be obtained from a faucet and is used to perform transactions on the Sepolia network. It does not have real monetary value, making it suitable for safely demonstrating blockchain transactions.

4. Cryptocurrency Transfer

A blockchain transaction involves a sender, recipient, amount, and network fee known as **gas**. To transfer cryptocurrency, the sender enters the recipient's wallet address and the amount to be transferred. MetaMask displays the transaction details and estimated gas fee before confirmation. After the user confirms and signs the transaction, it is broadcast to the blockchain network. Once the transaction is processed and confirmed, the recipient's balance increases and the sender's balance decreases by the transferred amount plus the applicable gas fee.

The basic workflow is:

MetaMask Account → Sepolia Testnet → Fund Account → Enter Recipient → Send 0.01 SepoliaETH → Confirm Transaction → Verify Transaction

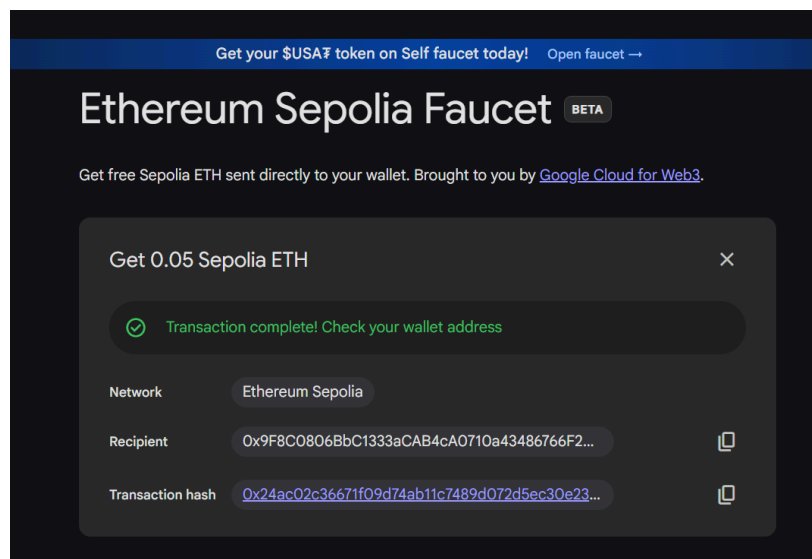
Procedure:

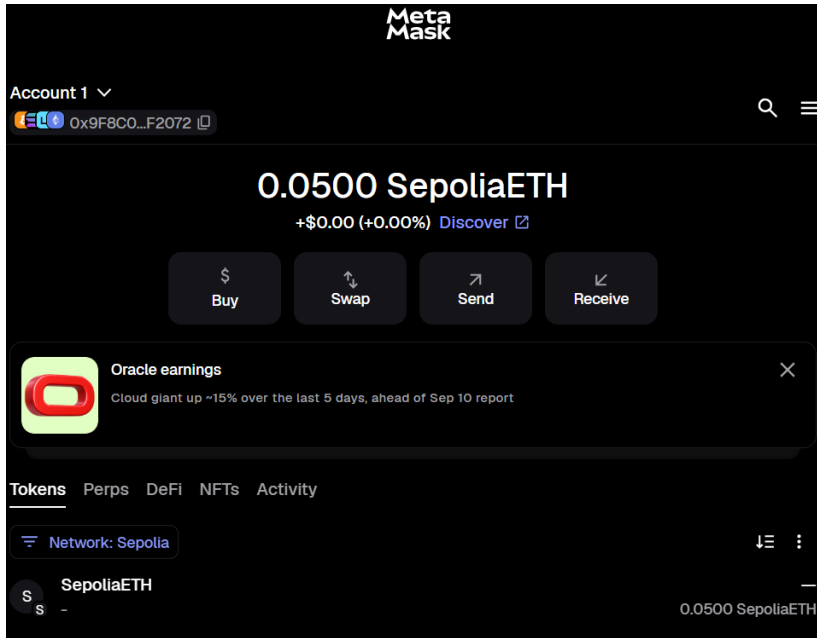
1. Open MetaMask and create or access a wallet account.
2. Configure and select the **Sepolia Testnet**.
3. Copy the wallet address of Account 1.
4. Obtain test SepoliaETH from a Sepolia faucet.
5. Verify that the test ETH has been received in Account 1.
6. Create a second MetaMask account, **Account 2**.
7. Copy the public wallet address of Account 2.
8. Switch back to Account 1.
9. Select the **Send** option in MetaMask.
10. Enter the Account 2 wallet address as the recipient.
11. Enter **0.01 SepoliaETH** as the amount.
12. Review the transaction details and estimated gas fee.
13. Confirm and sign the transaction.
14. Check the transaction status in the MetaMask Activity section.
15. Verify the updated balance of Account 2.
16. Verify the transaction on the Sepolia block explorer.

Output:

1. Account Created and Funded

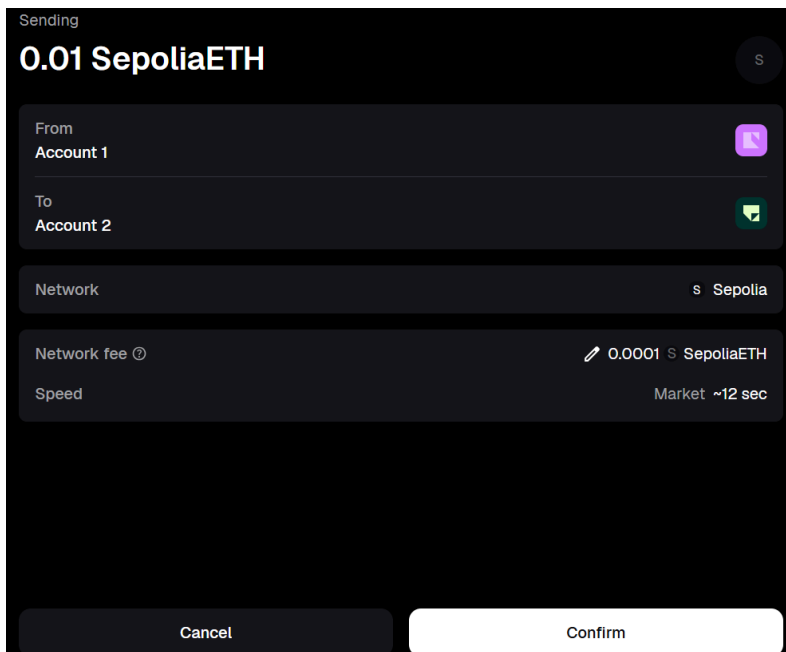
A MetaMask account was created and configured on the Sepolia Testnet. The account was successfully funded with **0.05 SepoliaETH** obtained from a testnet faucet.





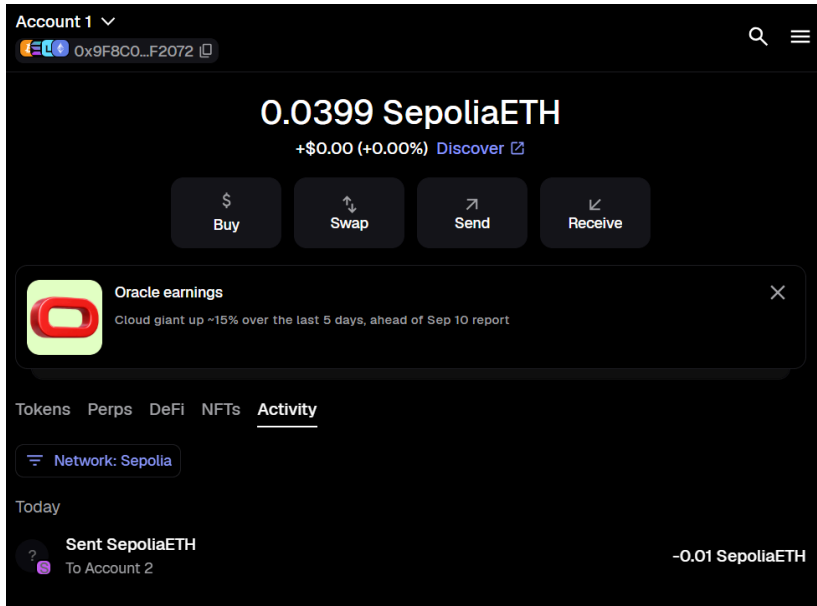
2. Initiating the Transfer

The **Send** option was selected in MetaMask. The recipient's Account 2 address was entered and **0.01 SepoliaETH** was specified for transfer.



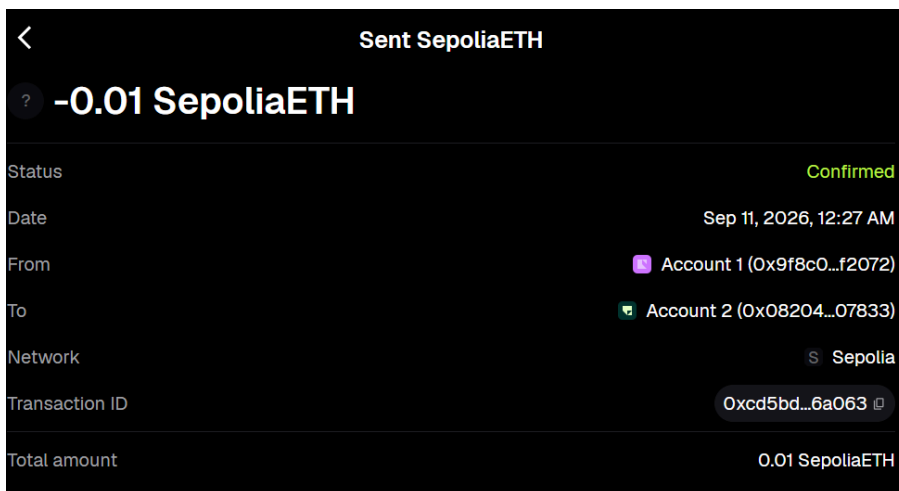
3. Confirming the Transaction

MetaMask displayed the transaction details including the sender, recipient, Sepolia network, transfer amount, and estimated network gas fee. The transaction was reviewed and confirmed.



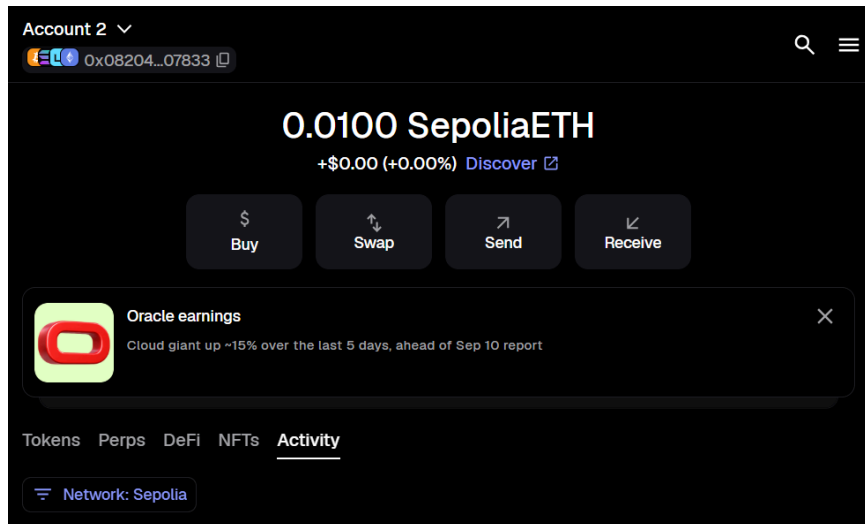
4. Transaction Confirmation

After confirmation, the transaction appeared in the MetaMask **Activity** section with a successful/confirmed status.



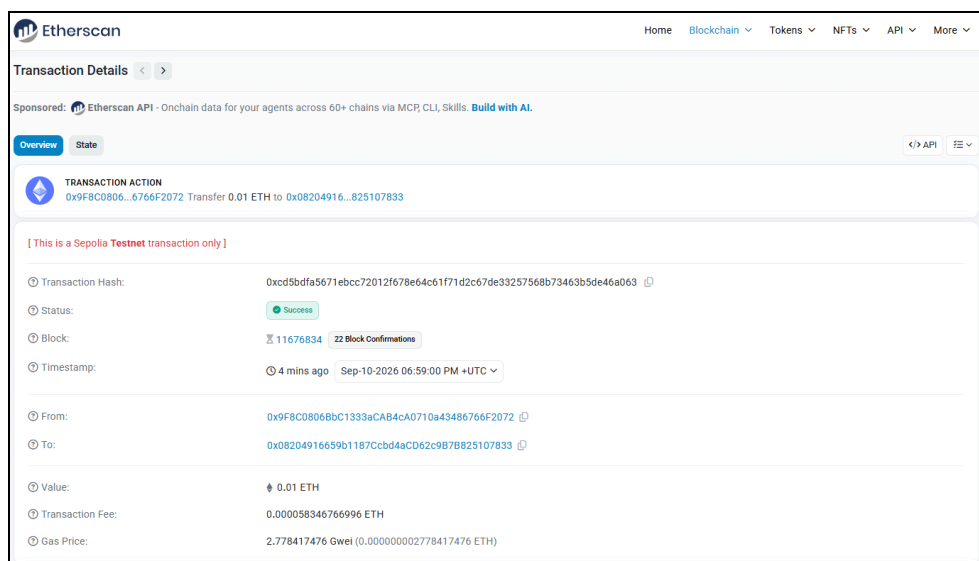
5. Verifying Updated Balance

The transaction was successfully processed and the balance of Account 2 was updated to reflect the received **0.01 SepoliaETH**.



6. Transaction on Block Explorer

The transaction was verified using the **Sepolia block explorer**. The transaction details show the sender address, recipient address, transferred value, transaction status, and transaction fee.



Conclusion:

In this experiment, a MetaMask wallet was created and configured on the Sepolia Testnet and funded with test SepoliaETH. A **0.01 SepoliaETH** transfer was made from Account 1 to Account 2 and successfully verified through MetaMask and the Sepolia block explorer. The experiment demonstrated wallet creation, cryptocurrency transfer, transaction verification, and the deduction of network gas fees.